

Bulk RNA-seq Differential Expression with DESeq2

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Introduction

Bulk RNA-seq asks which genes change in expression between conditions. DESeq2 is the most widely used framework for the analysis: it models counts with a negative binomial distribution, shares dispersion information across genes to stabilise estimates in small samples, and tests each gene's log-fold change for significance with multiple-testing correction. A standard DESeq2 workflow goes from a raw count matrix to a list of differentially expressed genes in a dozen lines of R.

Prerequisites

The reader should understand what a count matrix is (rows = genes, columns = samples), the basic biology of RNA-seq (reads mapped to transcripts, summed per gene), and the idea of a generalised linear model. Familiarity with Bioconductor data structures (`SummarizedExperiment`) is helpful.

Theory

DESeq2 assumes that the read count K_{ij} for gene i in sample j follows a negative binomial distribution

$$K_{ij} \sim \text{NB}(\text{mean} = \mu_{ij}, \text{dispersion} = \alpha_i),$$

where $\mu_{ij} = s_j q_{ij}$ is the product of a sample-specific size factor s_j (for sequencing depth normalisation) and a gene-specific expected count q_{ij} that depends on the experimental design through a log-linear model $\log_2 q_{ij} = \sum_k x_{jk} \beta_{ik}$.

Three ideas make DESeq2 work well on small-sample studies:

1. **Size factors** are computed by the median-of-ratios method, which is robust to highly expressed genes dominating the normalisation.
2. **Dispersion shrinkage** borrows strength across genes: each gene's dispersion estimate is shrunk toward a mean-dispersion trend fitted across all genes, so that noisy per-gene estimates in small samples do not produce wild inference.
3. **Log-fold-change shrinkage** (via `apeglm` or `ashr`) pulls LFCs of low-count genes toward zero, preventing them from dominating ranked gene lists.

The Wald test compares the estimated coefficient of interest to zero, with standard errors that reflect both the design and the dispersion. The Benjamini-Hochberg procedure controls the false discovery rate across the thousands of genes tested.

Assumptions

- Read counts follow a negative binomial distribution – usually reasonable for bulk mRNA-seq; may fail for very-low-coverage or heavily PCR-amplified libraries.
- The dispersion-mean trend is smooth across genes.
- Replicates within a condition are exchangeable after accounting for modelled covariates.
- No confounding between the biological condition and technical covariates (batch, library prep date). If there is, include the technical covariate in the design.

R Implementation

```
library(DESeq2)
library(apegglm)
library(ggplot2)

counts <- matrix(rnbinom(20000 * 6, mu = 100, size = 10),
                 nrow = 20000, ncol = 6)
rownames(counts) <- paste0("gene", seq_len(20000))
colnames(counts) <- paste0("sample", seq_len(6))
coldata <- data.frame(condition = factor(rep(c("control", "treated"), each = 3)),
                      row.names = colnames(counts))

counts[1:200, 4:6] <- counts[1:200, 4:6] * 3

dds <- DESeqDataSetFromMatrix(countData = counts,
                             colData = coldata,
                             design = ~ condition)

dds <- dds[rowSums(counts(dds) >= 10) >= 3, ]

dds <- DESeq(dds)

res <- lfcShrink(dds, coef = "condition_treated_vs_control",
               type = "apeglm")

summary(res)
head(res[order(res$padj), ])

plotMA(res, ylim = c(-3, 3))
```

The toy simulation above creates 20,000 genes and 6 samples, upregulates the first 200 genes in the treated condition, and runs the standard DESeq2 pipeline: construct the DESeqDataSet, filter very-low-count genes, estimate size factors and dispersion, fit the model, and shrink LFCs for reporting.

Output & Results

The `summary(res)` output reports the number of genes up- and down-regulated at the default alpha (0.1): typically a few hundred in the simulated upregulated set, with a small number of false positives at FDR 10%. The MA plot shows that the shrunk LFCs are compressed toward zero at low mean expression, as desired.

Interpretation

For a manuscript: “Differential expression between treated and control samples was assessed with DESeq2 (v1.44). Size factors were estimated by the median-of-ratios method; dispersions were shared across genes via the DESeq2 trend; log-fold changes were shrunk using `apeglm`. 212 genes were differentially expressed at an FDR of 5% (154 up, 58 down).”

Practical Tips

- Use `vst()` or `rlog()` transformations for downstream visualisation (PCA, heatmaps); use raw counts in DESeq2’s own pipeline.
- Filter low-count genes *before* running DESeq (e.g., keep genes with ≥ 10 counts in $\geq$ the smallest group size). This improves multiple-testing adjusted p-values without inflating false discoveries.

- Always inspect the PCA of normalised samples before differential expression. If samples cluster by batch rather than by condition, revise the design formula before interpreting results.
- `apeglm` shrinkage is the recommended default for LFC reporting. `ashr` handles contrasts; `normal` is the legacy method.
- Report shrunk LFCs for ranking and visualisation, but use the Wald test p-values for significance calling.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat